THE UNIVERSITY OF ALABAMA / CS 495

Capstone Design Manager

SPRINT 2 / PROGRESS PRESENTATION

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01 / THIS SPRINT

Less setup. Real deployment.

This sprint we improved team workflow, tested our changes, and got a hosted version running on Google Cloud.

01

Development setup

Get the application running locally with Docker.

02

Team workflow

Use GitHub issues, branches, and pull requests.

03

Public deployment

Host the app on Google Cloud with a public HTTPS address.

02 / WORK COMPLETED

What has changed since Sprint 1

01	**Production deployment**	Vue, Django, and PostgreSQL are running on a Google Cloud VM.
02	**GitHub collaboration**	Deployment PR #59 merged into main with three passing CI checks.
03	**Testing baseline**	API health check passed; an earlier local run passed 188 tests.

Remaining checks: user-role flows and file uploads.

03 / TECHNICAL OVERVIEW

How the application is put together

The hosted version runs on Google Cloud using Docker, with HTTPS in front of the application.

01

Vue + Nginx

Frontend

→

02

Django REST

API / Gunicorn

→

03

PostgreSQL

Database

Auth0

Login and user roles

Cloud Storage

File storage (verify uploads)

04 / GITHUB WORKFLOW

A clearer process for team changes

Instead of working in the same files and hoping everything merges, changes move through one repeatable process.

01 Choose an issue

02 Create a branch

03 Open a pull request

04 Run checks

05 Review + merge

Example: deployment PR #59 merged into main after three successful CI checks.

05 / QUALITY

Tests and health checks

3 / 3

CI checks passed

What we verified

Backend and frontend CI checks passed. The hosted API returned HTTP 200 with the database connected.

An earlier local run passed 188 tests. End-to-end user flows still need checking.

06 / MY WORK THIS SPRINT

Assignment solver and v1 cleanup

Two issues, two pull requests, both merged into main with passing CI.

#51 · PR #65

Proposed-assignment solver

The instructor runs OR-Tools on a semester, reviews the proposed teams, and approves. Nothing is saved before approval.

+556 lines · 22 new tests

#53 · PR #64

v1 cleanup

Pinned every backend package, recorded Python 3.11 and Node 24, and removed dead code left over from v1.

1,833 lines removed · 33 files

07 / HOW THE SOLVER WORKS

From rankings to proposed teams

Built on the OR-Tools CP-SAT spike; it runs inside Django admin, where the instructor already works.

01

Students rank

Five projects at each of ranks 1, 2, and 3.

02

Instructor runs Propose

Pick one semester; team size defaults to 4.

03

CP-SAT solves

Every student on one project, every team the right size.

04

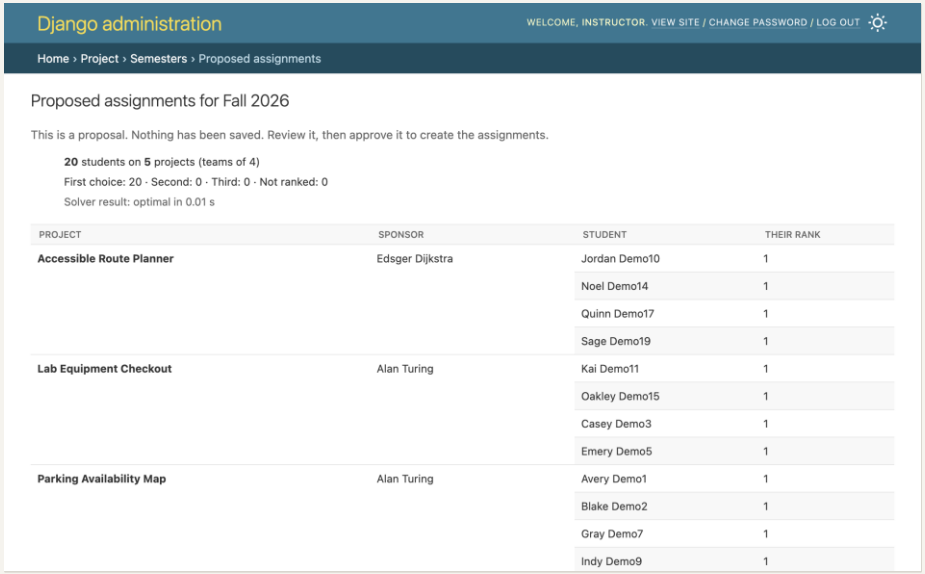
Review, then approve

Assignments are saved only on approval.

What it minimizes: 1st choice 0 · 2nd choice 1 · 3rd choice 3 · not ranked 10

08 / SOLVER DEMO

The proposal page



Django admin, demo data (seed_demo), Fall 2026

0.01 s

to solve 20 students and 16 projects

20 / 20

students got a first choice

0 rows

saved until the instructor approves

09 / QUALITY + CLEANUP

More tests, less dead code

258

backend tests passing, 22 of them new for the solver

28

frontend tests; 8 went with the dead components they tested

01 Pinned versions

Every backend package pinned. CI now uses Python 3.11 and Node 24, the same as Docker.

02 Removed dead code

Six unused Vue components, unused email views, an unmounted admin site, and unused templates.

03 Settings, not constants

The email year and the cloud storage project now come from settings.

10 / CHALLENGES

The parts that took extra work

Getting services online

We fixed Docker permissions, connected PostgreSQL, and set up the public HTTPS reverse proxy.

Switching authentication

We moved to the team Auth0 tenant and resolved the callback, client ID, and API access issues.

11 / DEMO READINESS

The website is live

A hosted version is running independently of our own computers, ready for a live demo.

VERIFIED THIS SPRINT

- Public HTTPS site on Google Cloud
- API HTTP 200; PostgreSQL connected

STILL TO VERIFY

- Student and sponsor dashboards
- Auth0 roles and complete user flow
- Upload and download functionality

ua-capstone-design.duckdns.org | Deployment PR #59 merged

12 / NEXT SPRINT

Where we go from here

01	**Finish end-to-end tests**	Confirm both dashboards, account roles, and file uploads.
02	**Work through user flows**	Sponsor submissions, student preferences, and instructor tools.
03	**Document + improve**	Deployment handoff, clearer errors, more tests, and sponsor feedback.

Demo & questions

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github.com/monicadelaine/capstone-design